INCH-POUND

MIL-DTL-85097/13C(OS)
20 April 2000
SUPERSEDING
MIL-DTL-85097/13B(OS)
1 March 1999

DETAIL SPECIFICATION

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR, CANOPY JETTISON, MARK 109 MODS 0 AND 1 (LOADED ASSEMBLIES)

This specification sheet is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements unique to the Aircrew Escape Propulsion Systems (AEPS) Canopy Jettison Rocket Motor Mark 109 Mods 0 and 1 Loaded Assemblies (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-P-83126 Propulsion Systems, Aircrew Escape, Design Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Team (Code 840M), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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MIL-DTL-85097 Aircrew Escape Propulsion Systems (AEPS), General Specification For

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-464 Electromagnetic Environmental Effects Requirements for Systems

(Unless otherwise indicated, copies are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

NAVAL SURFACE WARFARE CENTER INDIAN HEAD DIVISION (CAGE Code 14083)

510-174TD-004 Propellant, NSWC/IH-EC-8

511-174TD-004 Propellant, NOSIH-BC-11, NSWC/IH-EC-02, NSWC/IH-EC-04, and NSWC/IH-EC-05

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.2.2 Other Government documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL1507AS100 Rocket Motor, Canopy Jettison, Mk 109 Mod 0

DL1507AS201 Rocket Motor, Canopy Jettison, Mk 109 Mod 1

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-DTL-85097.

3.1.1 Design. The Rocket Motors described herein shall conform to DL1507AS100 (Mk 109 Mod 0) or DL1507AS201 (Mk 109 Mod 1).

3.2 Propellant aging. When tested in accordance with MIL-DTL-85097, the propellant shall meet the aging requirements of 511-174TD-004 (Mk 109 Mod 0) or 510-174TD-004 (Mk 109 Mod 1).

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3.3 Propellant. The propellant shall conform to the requirements of 511-174TD-004 (Mk 109 Mod 0) or 510-174TD-004 (Mk 109 Mod 1).

3.4 Ballistic performance requirements. The rocket motors shall meet the ballistic performance requirements of table I when test fired at both −65 ± 5°F and 165 ± 5°F.

TABLE I. Ballistic performance requirements.

Parameter	−65°		+165°	
	Min	Max	Min	Max
Total impulse (lb-sec)	178	211	181	216
Action time (ms)	125	172	88	143
Max Thrust (lb)	1305	1933	1665	2746
Thrust Angle (deg)	86	94	86	94
Time to 70% Max Thrust (ms)	-	20	-	20

3.5 Hydrostatic proof test. Each motor tube shall meet the hydrostatic test requirements of drawing 1507AS132.

3.6 Leakage. Each rocket motor assembly shall have a leakage rate of less than 1×10^{-4} cc/sec air when subjected to the dry leak test of 4.1.6.

3.7 Threaded joint torque. The 1.890-24NS-3 threaded joints (2) of each rocket motor assembly shall withstand 400-inch-pounds torque without breakaway, when tested in accordance with MIL-P-83126.

3.8 Electrical bonding resistance. The electrical bonding resistance between the shielded mild detonating cord (SMDC) housing and the mounting lugs of each rocket motor assembly shall be no greater than 0.0025 ohms, when tested in accordance with MIL-STD-464.

3.9 Motor weight. The weight of the rocket motor shall not exceed 3.50 lbs.

4. VERIFICATION

4.1 First article/quality conformance. First article and quality conformance tests shall be performed as specified in MIL-DTL-85097 with the conditions specified in 4.1.1 through 4.1.8.

4.1.1 Propellant test results. The contractor shall submit propellant test results in accordance with MIL-DTL-85097 and 511-174TD-004 (Mk 109 Mod 0) or MIL-DTL-85097 and 510-174TD-004 (Mk 109 Mod 1).

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4.1.2 Conditioning temperature. The conditioning temperatures and tolerances shall be as follows:

a. Cold: −65 ± 5°F
b. Hot: 165 ± 5°F

4.1.3 Conditioning period. Time for temperature conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: 4 hours, minimum
b. Temperature cycling: 5 hours, minimum

4.1.4 Radiographic inspection. All components or assemblies shall be radiographed in two views 90° apart.

4.1.5 Hydrostatic proof test. Each motor tube shall be subjected to the hydrostatic tests of drawing 1507AS132.

4.1.6 Leakage test. Each rocket motor assembly shall be subjected to a dry leak test to verify compliance with a leakage rate of less than 1×10^{-4} cc/sec air.

4.1.7 Threaded joint torque test. The 1.890-24NS-3 threaded joints (2) of each rocket motor assembly shall be subjected to the torque test of MIL-P-83126.

4.1.8 Electrical bonding resistance test. The electrical bonding resistance between the shielded mild detonating cord (SMDC) housing and the mounting lugs of each rocket motor assembly, selected as a sampling test specimen, shall be tested in accordance with MIL-STD-464.

5. PACKAGING

(This section is not applicable to this specification.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rocket motors described herein are intended to be used to eject the canopy during an emergency crew ejection from the F/A/TF-18 Aircraft.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2).

c. Mark and Mod designation of rocket motor.

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6.3 Definitions.

6.3.1 Maximum thrust. Maximum thrust is defined as the maximum recorded thrust that occurs over the action time.

6.3.2 Time to 70% maximum thrust. Time to 70% maximum thrust is defined as the elapsed time from the application of an electric current to the blasting cap (test tip) until 70% of the maximum thrust.

6.3.3 Impulse. Impulse is obtained by integrating the thrust of the rocket motor over the action time.

6.3.4 Action time. Action time is defined as the elapsed time from 10% of the maximum thrust on the initial rise of the thrust-time curve to the corresponding 10% of maximum thrust on the declining portion of the curve.

6.3.5 Thrust angle. Thrust angle is defined as the time averaged angle of the thrust vector as measured in the time interval between 10 percent of maximum thrust on the initial rise of the curve and the corresponding 10 percent of maximum thrust on the final declining portion of the curve on the rocket thrust trace.

6.4 Subject term (key word) listing.

Ballistic performance
Detonating cord
Propellant
Radiographic inspection

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - OS
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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-85097/13	2. DOCUMENT DATE (YYMMDD) 000420

3. DOCUMENT TITLE
AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR, CANOPY JETTISON, MARK 109 MODS 0 AND 1 (LOADED ASSEMBLIES)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700 (2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Standardization Team (Code 840M) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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